



EUROPEAN  
COMMISSION

Brussels, XXX  
[...] (2026) XXX draft

ANNEX

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**to the**

**Commission Delegated Regulation**

**supplementing Regulation (EU) 2024/3012 of the European Parliament and of the  
Council by establishing the certification methodologies for carbon farming activities**



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## ANNEX

### **CERTIFICATION METHODOLOGIES REFERRED TO IN ARTICLES 1, 2 AND 3**

#### **DEFINITIONS**

For the purposes of this Annex, the following definitions shall apply:

- (1) ‘eligible activity’ means an activity that is eligible for certification under Regulation (EU) 2024/3012;
- (2) ‘organic soil’ means organic soil as established in accordance with the 2006 Intergovernmental Panel on Climate Change (IPCC) Guidelines<sup>1</sup> (‘2006 IPCC Guidelines’) or under national definitions accepted under greenhouse gas (GHG) inventories reporting. Soils that are not organic are mineral soils;
- (3) ‘water table’ means the height of a phreatic water surface relative to the soil surface;
- (4) ‘activity area’ means the spatially delimited area or areas where the activity takes place;
- (5) ‘peat’ means a layer of soil that primarily consists of partially decomposed plant material, that has accumulated under conditions of waterlogging;
- (6) ‘certification period’ means the period between a re-certification audit of an activity and the most recent preceding certification audit or re-certification audit of that activity;
- (7) ‘renewal’ means the start of a new activity period;
- (8) ‘peat depletion time’ means a time period over which the total peat layer would disappear in the absence of the activity;
- (9) ‘calibration’ means a process involving the adjustment of parameters and constants within a model so that it predicts the measured values more accurately;
- (10) ‘validation’ means the process of evaluating the performance of a model with regard to measured values, demonstrating its satisfactory performance in terms of suitability and characterisation of model prediction error;
- (11) ‘natural disturbance event’ means an event or series of events resulting from natural processes, such as wildfires, storms, extreme climate events, or biological and insect outbreaks, that lead to significant alterations in the structure and functioning of ecosystems.

#### **1. DESCRIPTION OF THE CARBON FARMING ACTIVITY**

##### **1.1. Eligibility criteria**

###### *1.1.1. Agriculture and agroforestry on mineral soils*

###### **1.1.1.1. Eligible activity**

The eligible activity shall be carried out on mineral agricultural soils (cropland and grassland) and may cover the following practices:

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<sup>1</sup> 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4 ‘Agriculture, Forestry and Other Land Use’, Chapter 3 ‘Default climate and soil classifications’, Annex 3A.5 .

- (a) agricultural practices that increase net carbon removals in soils or reduce net CO<sub>2</sub> emissions from soils, such as:
  - (i) improved crop management that increases soil cover and/or that increases the amount of carbon input from crop residues to the soil such as cover crops, crop rotations, crop residue retention;
  - (ii) conservation tillage practices that reduce soil disturbances;
  - (iii) conversion of cropland to grassland;
  - (iv) improved grassland management such as conversion of arable land for fodder crops to permanent grassland, rotational grazing, mixed swards;
  - (v) use of organic soil improvers or organic fertilisers;
- (b) agroforestry practices that increase net carbon removals in living biomass, such as:
  - (i) tree planting inside parcels such as silvopastoral or silvoarable agroforestry;
  - (ii) planting of woody elements between parcels such as hedges or other landscape features;
  - (iii) new permanent crops where two or more agricultural practices are carried out such as grazed or intercropped orchards;
- (c) agricultural and agroforestry practices that reduce direct and indirect N<sub>2</sub>O emissions from managed agricultural soils, such as:
  - (i) precision fertilisation;
  - (ii) replacement of mineral nitrogen fertilisers by growing leguminous crops, using soil improvers or plant biostimulants;
  - (iii) change in type of fertilising products;
  - (iv) use of nitrification, denitrification or urease inhibitors.

Several practices may be combined within the activity area.

#### 1.1.1.2. Eligibility requirements

The following eligibility requirements shall apply:

- (a) conservation tillage practices are eligible only if combined with practices over the rotation that increase the carbon input into the soils;
- (b) conversion of cropland to grassland shall not take place on land where permanent grassland was converted to cropland after 1 January 2023;
- (c) the activity shall not result in the removal of existing trees, hedges or other carbon-rich landscape features;
- (d) improved grassland management shall not take place on land where permanent grassland was ploughed after 1 January 2023;
- (e) the activity shall not take place on land where the water table has been artificially lowered after 1 January 2023;

- (f) biochar may be applied only where the quantification approach allows to exclude from the quantification of soil organic carbon stocks the amount of carbon applied through biochar eligible under Commission Delegated Regulation [...]²;
- (g) the practices referred to in Section 1.1.1.1., point (b), shall not take place on land where existing agroforestry systems were removed after 1 January 2023, save where this occurred for the purposes of control of fire, pest or windthrow events;
- (h) the practices referred to in Section 1.1.1.1., point (c), are eligible only if they complement the practices referred to in point (a) or point (b) of that Section and shall aim to improve the nitrogen use efficiency of crop production, that is, the ratio between the nitrogen crop output and the nitrogen inputs, and to maintain crop production levels on the activity area.
- (i) the activity shall maintain riparian buffer zones, such as riparian forests, buffer strips, meadows or pastures, along streams in the activity area, in accordance with Directive 2009/128/EC of the European Parliament and of the Council³.

For the purposes of the the practices referred to in Section 1.1.1.1., point (b), management practices shall be carried out in a way that minimises negative impacts on soil quality and soil health. Tree species qualifying as unsuitable for current and future climate conditions as a result of the risk assessment carried out under Section 4.1.3 shall not be eligible.

### 1.1.2. *Rewetting and restoration of peatlands and of other organic soils*

#### 1.1.2.1. Eligible activity

The eligible activity shall result in a rise of the water table in organic soils and peatlands and may cover the following practices:

- (a) blocking, damming, infilling or removing of drainage structures;
- (b) reducing or stopping local water extraction, creating or facilitating aboveground structures that hamper rainwater surface run-off, and decreasing groundwater extraction.

The practices referred to in the first paragraph, points (a) and (b), may be combined with:

- (a) the active re-establishment of peat-forming vegetation and shallow removal of degraded topsoil or atypical vegetation;
- (b) the application of paludiculture.

An activity qualifying as high-risk as a result of the risk assessment carried out under Section 4.1.2 shall not be eligible.

#### 1.1.2.2. Eligibility requirements

The following eligibility requirements shall apply:

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<sup>2</sup> Commission Delegated Regulation [...] supplementing Regulation (EU) 2024/3012 of the European Parliament and of the Council by establishing the certification methodologies for permanent carbon removals activities.

<sup>3</sup> Directive 2009/128/EC of the European Parliament and of the Council of 21 October 2009 establishing a framework for Community action to achieve the sustainable use of pesticides (OJ L 309, 24.11.2009, p. 71, ELI: <http://data.europa.eu/eli/dir/2009/128/oj>).

- (a) the activity shall not take place on land where the water table was artificially lowered after 1 January 2023;
- (b) mineral soils may be included in the activity area if the changes in emissions on those soils as a result of the activity are accounted for;
- (c) no peat extraction and burning of peat shall take place in the activity area;
- (d) for paludiculture the following shall apply:
  - (i) soil disturbance shall be minimised when seeding, managing or harvesting, in line with best available practices;
  - (ii) below-ground biomass shall not be harvested and mosses may only be harvested from sites where they are actively cultivated.

### 1.1.3. *Afforestation*

#### 1.1.3.1. Eligible activity

The eligible activity shall result in the afforestation of:

- (a) grassland;
- (b) cropland;
- (c) other non-forest land.

The activity may include direct planting and seeding and practices that enable or facilitate natural regeneration.

#### 1.1.3.2. Eligibility requirements

The following eligibility requirements shall apply:

- (a) the activity shall not take place on land where, in the 20 years prior to the start of the activity period, the combined tree crown cover of existing sparse trees is larger than 10 % of the activity area; those trees shall not be removed at the start of the activity, save where they are included in the list of invasive alien species of Union concern pursuant to Commission Implementing Regulation (EU) 2016/1141<sup>4</sup>;
- (b) the minimum size of the activity area shall be 0,5 ha;
- (c) the activity may take place on peatlands, only if they are subject to rewetting;
- (d) on organic soils other than peatlands, the activity shall not result in the lowering of the water table or in any form of soil degradation;
- (e) the activity shall reflect the projected age structure and species composition that is typical of the local natural conditions determined by soil, hydrological and climate conditions and specific local factors like wind exposure;
- (f) mixed species composition shall be the common approach, whereby a single species may be applied in cases where this reflects natural species composition and this is the most appropriate species adapted to climate and soil conditions;

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<sup>4</sup> Commission Implementing Regulation (EU) 2016/1141 of 13 July 2016 adopting a list of invasive alien species of Union concern pursuant to Regulation (EU) No 1143/2014 of the European Parliament and of the Council (OJ L 189, 14.7.2016, p. 4, ELI: [http://data.europa.eu/eli/reg\\_impl/2016/1141/oj](http://data.europa.eu/eli/reg_impl/2016/1141/oj)).

- (g) with the exception of natural regeneration, the planting density throughout the activity period shall be in line with planting densities or with equivalent requirements applicable in the Member State where the activity is implemented;
- (h) management practices shall be carried out in a way that minimises negative impacts on soil quality and soil health. In particular:
  - (i) tillage shall be allowed only for the purpose of site preparation and planting/seeding, either at the start of the activity or if trees are planted at later stages to recover losses from failure or harvest;
  - (ii) mowing shall be allowed only to the extent and duration that is necessary to provide growth space for the planted trees and for the purposes of fire prevention.

For the purposes of the first paragraph, point (e), tree species qualifying as unsuitable for current and future climate conditions as a result of the risk assessment carried out in accordance with Section 4.1.3 shall not be eligible, except for species for which historical or experimental success as well as potential future suitability in the region is demonstrated in the activity plan referred to in Section 1.3.1. Native species shall be used, whereas some non-native species may be used where it is demonstrated that the native species and their provenances are no longer adapted to the projected climatic and pedo-hydrological conditions of the activity area and the use of the forest reproductive material leads to improved ecosystem conditions.

For the purposes of the first paragraph, point (f), where the initial growing conditions have improved, operators shall introduce additional species, including through natural establishment, that shall be subject to the conditions laid down in the first paragraph, point (e).

For the purposes of the first paragraph, point (h)(i), tillage shall not exceed 15 cm of depth and where afforestation takes place on grassland, no tillage shall be applied. Where this is not possible due to local soil conditions, the relevant justification shall be included in the activity plan.

## **1.2. Activity period, monitoring period and certification period**

### **1.2.1. Agriculture and agroforestry on mineral soils**

#### **1.2.1.1. Activity period**

The duration of the activity period shall be:

- (a) for the practices referred to in Section 1.1.1.1, points (a) and (c): five years, renewable up to three times for periods of the same duration;
- (b) for the practices referred to in Section 1.1.1.1, point (b): 15 years, renewable once for a period of the same duration.

At the time of submission of the activity plan referred to in Section 1.3.1, operators or groups of operators may already commit, to an overall activity period of maximum 20 years for the practices referred to in Section 1.1.1.1, point (a), or of maximum 30 years for the practices referred to in Section 1.1.1.1, point (b).

#### **1.2.1.2. Monitoring period**

The monitoring period shall start at the same time as the activity period, and its duration shall be at least:

- (a) 10 years longer than the activity period: for the practices referred to in Section 1.1.1.1, points (a) and (b);
- (b) equivalent to the duration of the activity period: for the practices referred to in Section 1.1.1.1, point (c).

Operators or groups of operators may already commit to a longer monitoring period at the time of submission of the activity plan.

#### *1.2.2. Rewetting and restoration of peatlands and of other organic soils*

##### *1.2.2.1. Activity period*

The duration of the activity period shall be at least 10 years, renewable up to a maximum overall activity period of 30 years.

The duration of the activity period, including renewals, shall not exceed the maximum peat depletion time identified in the entire activity area in the absence of the activity.

The peat depletion rate shall be:

- (a) 1.0 cm per annum on peatlands with acidic, nutrient-poor conditions primarily receiving water from precipitation, such as bogs;
- (b) 1.5 cm per annum on nutrient-rich peatlands fed by ground water or surface water, such as fens.

These figures shall be increased in the case of peat erosion by water, wind or frost action and may be decreased where operators or groups of operators demonstrate with a scientific rationale that the rate is lower, e.g. in shallowly drained peatland.

Where subsidence occurs, the determination of the relevant peat depletion rates shall be supported by reliable and verifiable information based on expert judgment, relevant datasets, or scientific literature pertaining to the activity area or comparable regions.

The activity area shall be stratified in strata or parcels with comparable peat depths and peat depletion times before the beginning of the activity period.

Peat depth measurements shall be taken in points to be selected by using a systematic or randomly stratified raster or transects with appropriate cover and spacing to allow for an adequate and comprehensive determination of the relevant peat depths over the entire activity area.

In the case of relevant height differences within the activity area, the selection of observation points shall be guided by a high-resolution digital elevation model.

Peat depth measurements shall be cored up to a depth of maximum 50 cm. The coring shall follow standard scientific protocols, and the method of choice shall be justified in the activity plan.

The first re-certification audit shall be carried out during the fifth year from the start of the activity.

By way of derogation from the previous paragraph, the first re-certification audit may be carried out during the first four years of the activity, if the absence of a methane emission peak is demonstrated. For the cases referred to in Section 3.2.1, third paragraph, points (a) and (b), the demonstration of the absence of a methane peak is not required, where the activity started five years before the submission of the activity plan or the recognition of the certification scheme.

#### 1.2.2.2. Monitoring period

The duration of the monitoring period shall correspond to the duration of the activity period.

### 1.2.3. *Afforestation*

#### 1.2.3.1. Activity period

The duration of the activity period shall be 30 years.

No renewal is allowed.

#### 1.2.3.2. Monitoring period

The monitoring period shall start at the same time as the activity period and its duration shall be at least 40 years.

Operators or groups of operators may already commit to a longer monitoring period at the time of submission of the activity plan.

### 1.3. **Planning and reporting**

Before every re-certification audit, operators or groups of operators shall submit to the certification body a monitoring report including the information specified in Section 1.3.4.1.

Before every monitoring audit, operators or groups of operators shall submit to the certification body a monitoring report including the information specified in Section 1.3.4.2.

In the event of renewal of the activity, Article 9(1), second subparagraph, of Regulation (EU) 2024/3012 shall apply.

#### 1.3.1. *Activity plan*

The activity plan shall include the following:

- (a) general information:
  - (i) legal ownership and contact information;
  - (ii) georeferenced boundaries of the activity area, including, where applicable, codes from the national Integrated Administration and Control System or Land Parcel Identification System (LPIS), and planned size of the activity area;
  - (iii) duration of the activity period, including the start date;
  - (iv) the information referred in Article 8(1) of Commission Implementing Regulation (EU) 2025/2358<sup>5</sup>;
- (b) with regard to the eligibility requirements:
  - (i) description of intended practices;
  - (ii) evidence of previous land use, based on sources such as the LPIS, recent measurements or remote sensing data;
- (c) with regard to the quantification criterion:

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<sup>5</sup> Commission Implementing Regulation (EU) 2025/2358 of 20 November 2025 laying down rules on certification schemes, certification bodies, and audits under Regulation (EU) 2024/3012 of the European Parliament and of the Council (OJ L, 2025/2358, 21.11.2025, ELI: [http://data.europa.eu/eli/reg\\_impl/2025/2358/oj](http://data.europa.eu/eli/reg_impl/2025/2358/oj)).

- (i) the estimated baseline, including the assumptions used to calculate it, broken down by carbon removals and soil emissions;
  - (ii) the estimated carbon removals and/or soil emissions to be generated by the activity, broken down by total carbon removals, total soil emissions, and other GHG emissions associated to the activity;
- (d) with regard to the additionality criterion:
  - (i) regulatory test;
  - (ii) incentive effect test;
  - (iii) financial viability test;
- (e) with regard to the storage, monitoring, and liability criterion:
  - (i) assessment of reversal risks;
  - (ii) proposed practices to mitigate reversal risks;
  - (iii) chosen liability mechanism;
- (f) with regard to the sustainability criterion:
  - (i) description of how the activity complies with the minimum sustainability requirements referred to in Section 5.1;
  - (ii) a description of how the activity complies with the obligation to generate co-benefits for the protection and restoration of biodiversity and ecosystems, including soil health and the avoidance of land degradation;
  - (iii) where applicable, expected sustainability co-benefits;
- (g) a monitoring plan as specified in Section 1.3.2.

#### *1.3.2. Monitoring plan*

The monitoring plan shall consist of detailed, complete and transparent documentation about how operators or groups of operators plan to monitor the activity. Certification schemes may provide additional guidance specifying which elements shall be included for each type of activity, more frequent measurement requirements and/or more detailed requirements for quality assurance.

The monitoring plan shall include the following:

- (a) a description of the carbon pools and emission sources referred to in Tables 1 and 2 that will be used for quantification and, for each of them, the chosen quantification or monitoring approach and the frequency of monitoring;
- (b) the relevant documentation listed in Sections 2.4.1.3, 2.4.2.3 and 2.4.3.3;
- (c) a description of how the risks of reversals and the reversals, should they occur, will be monitored throughout the monitoring period;
- (d) a description of the approach to monitor co-benefits for the sustainability objectives throughout the activity period.

#### *1.3.3. Update of the activity or monitoring plan*

In the event of permanent changes to any of the elements listed in Sections 1.3.1 and 1.3.2, operators or groups of operators shall submit a revised activity plan and/or monitoring plan

describing the nature, extent of, and the rationale of the proposed or actual changes. In the revised activity or monitoring plan, operators or groups of operators shall report any impacts of the changes on the compliance with the applicable methodology. Temporary changes shall be described only in the monitoring report.

#### 1.3.4. Monitoring report

Operators or groups of operators shall obtain, record, compile, analyse and document monitoring data, including assumptions, references, activity data and calculation factors in a transparent manner that enables the checking of performance achieved during the certification period, when requested, and report this information to the certification bodies or certification schemes.

Certification schemes shall allow for the sharing of monitoring data in an anonymised way with the Commission upon request by competent national authorities or the Commission.

##### 1.3.4.1. Monitoring report before re-certification audits

For the purposes of the re-certification audits, the monitoring report shall include:

- (a) the information listed in Table 1 and Table 2, where relevant;

Table 1

| LULUCF carbon pools and LULUCF emission sources                       | GHG                                                  | Data unit                 | Description                   | Applicable to                                                                    |
|-----------------------------------------------------------------------|------------------------------------------------------|---------------------------|-------------------------------|----------------------------------------------------------------------------------|
| Living biomass                                                        | CO <sub>2</sub>                                      | tonnes CO <sub>2</sub> eq | Amount of total net removals  | Agriculture and agroforestry on mineral soils, afforestation                     |
|                                                                       |                                                      |                           | Amount of total net emissions | Rewetting and restoration of peatlands and of other organic soils                |
| Mineral soils                                                         | CO <sub>2</sub>                                      | tonnes CO <sub>2</sub> eq | Amount of total net removals  | Agriculture and agroforestry on mineral soils, afforestation                     |
|                                                                       |                                                      |                           | Amount of total net emissions | Rewetting and restoration of peatlands and of other organic soils                |
| Organic soils                                                         | CO <sub>2</sub> , CH <sub>4</sub> , N <sub>2</sub> O | tonnes CO <sub>2</sub> eq | Amount of total net emissions | Rewetting and restoration of peatlands and of other organic soils                |
| Direct N <sub>2</sub> O emissions from managed non-agricultural soils | N <sub>2</sub> O                                     | tonnes CO <sub>2</sub> eq | Amount of total net emissions | Rewetting and restoration of peatlands and of other organic soils, afforestation |

Table 2

| Non-LULUCF emission sources                                                    | GHG                                                  | Data unit                 | Description               | Applicable to                                                                                                                   |
|--------------------------------------------------------------------------------|------------------------------------------------------|---------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Managed agricultural soils                                                     | N <sub>2</sub> O                                     | tonnes CO <sub>2</sub> eq | Amount of total emissions | Agriculture and agroforestry on mineral soils, rewetting and restoration of peatlands and of other organic soils, afforestation |
| Managed agricultural soils (organic soils emissions in cropland and grassland) | N <sub>2</sub> O                                     | tonnes CO <sub>2</sub> eq | Amount of total emissions | Rewetting and restoration of peatlands and of other organic soils, afforestation                                                |
| Liming application                                                             | CO <sub>2</sub>                                      | tonnes CO <sub>2</sub> eq | Amount of total emissions | Agriculture and agroforestry on mineral soils, rewetting and restoration of peatlands and of other organic soils                |
| Urea application                                                               | CO <sub>2</sub>                                      | tonnes CO <sub>2</sub> eq | Amount of total emissions | Agriculture and agroforestry on mineral soils, rewetting and restoration of peatlands and of other organic soils                |
| Fuel combustion                                                                | CO <sub>2</sub> , CH <sub>4</sub> , N <sub>2</sub> O | tonnes CO <sub>2</sub> eq | Amount of total emissions | Agriculture and agroforestry on mineral soils, rewetting and restoration of peatlands and of other organic soils, afforestation |

- (b) information about the use of the relevant quantification and monitoring approaches laid down in Sections 2.4.1.2, 2.4.2.2 and 2.4.3.2, and their estimated uncertainties;
- (c) information to verify compliance with the additionality tests laid down in Section 3;
- (d) information about any occurred reversal referred to in Section 1.3.4.2, second paragraph;
- (e) information to verify compliance with the relevant sustainability requirements laid down in Section 5.

The monitoring report shall include the information referred to in points (c) and (e) only every five years.

#### 1.3.4.2. Monitoring report before monitoring audits

For the purposes of the monitoring audits, operators or groups of operators shall submit the monitoring report at least every five years and within one year of becoming aware of an event resulting in a reversal.

If a reversal has occurred, the monitoring report shall:

- (a) include a description of the monitoring approach used;
- (b) identify the location of the event in a geospatial vector format, such as shapefile, Keyhole Markup Language or similar formats, as one or more polygons, or by

specifying the coordinates of the geographic boundary using a standard coordinate system, such as ETRS89 or WGS84;

- (c) include a description and relevant data related to the observed reversal event and classify the event as being avoidable or unavoidable;
- (d) quantify the amount of the reversal by using the quantification rules laid down in Section 2 or, where this is not possible due to the nature of the reversal event, produce a conservative estimate of the reversal and evidence why the estimate is conservative;
- (e) describe the measures undertaken to stop the reversal and to prevent or minimise similar events occurring in the future.

## 2. QUANTIFICATION OF BASELINE, TOTAL CARBON REMOVALS, LULUCF SOIL EMISSIONS, $GHG_{\text{ASSOCIATED}}$ , AND UNCERTAINTY DEDUCTION FACTOR

### 2.1. Scope

The temporary net carbon removal benefit and the net soil emission reduction benefit shall be quantified in accordance with equations (1) and (2).

|                                                                                                                                                                                           |                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <p>Temporary net carbon removal benefit =</p> $(CR_{\text{baseline}} - CR_{\text{activity}}) \times (1 - UNC) - GHG_{\text{associated}}$                                                  | <div style="border-left: 1px solid black; padding-left: 10px;">equation<br/>(1)</div> |
| <p>Net soil emission reduction benefit</p> $= (LSE_{\text{baseline}} - LSE_{\text{activity}} + ASE_{\text{baseline}} - ASE_{\text{activity}}) \times (1 - UNC) - GHG_{\text{associated}}$ | <div style="border-left: 1px solid black; padding-left: 10px;">equation<br/>(2)</div> |

Where:

- (subscript) baseline = the quantity corresponding to the absence of the activity ('baseline scenario')
- (subscript) activity = the quantity corresponding to the implementation of the activity ('activity scenario')
- UNC = the factor representing the uncertainty at the level of the group of operators which shall be deducted to obtain a conservative estimate of total carbon removals or emission reductions ('uncertainty deduction factor').

Table 3 shows the correspondence between the terms of equations (1) and (2) and the LULUCF carbon pools and GHG emission sources defined in the 2006 IPCC Guidelines.

Table 3

| IPCC category | Term                     | IPCC carbon pool or GHG emission source    | GHG             |
|---------------|--------------------------|--------------------------------------------|-----------------|
| LULUCF        | CR (net carbon removals) | Living biomass                             | CO <sub>2</sub> |
|               |                          | Soil Organic Carbon (SOC) in mineral soils | CO <sub>2</sub> |

|                   |                                         |                                                                                |                                                      |
|-------------------|-----------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------|
|                   | <b>LSE (net LULUCF Soil Emissions )</b> | Soil Organic Carbon (SOC) in mineral soils                                     | CO <sub>2</sub> , N <sub>2</sub> O                   |
|                   |                                         | Soil Organic Carbon (SOC) in organic soils                                     | CO <sub>2</sub> , CH <sub>4</sub> , N <sub>2</sub> O |
| <b>Non-LULUCF</b> | <b>ASE (Agriculture Soil Emissions)</b> | Direct and indirect N <sub>2</sub> O emissions from managed agricultural soils | N <sub>2</sub> O                                     |
|                   |                                         |                                                                                |                                                      |
|                   | <b>GHG<sub>associated</sub></b>         | Living biomass                                                                 | CO <sub>2</sub>                                      |
|                   |                                         | Direct and indirect N <sub>2</sub> O emissions from managed agricultural soils | N <sub>2</sub> O                                     |
|                   |                                         | Liming & urea application                                                      | CO <sub>2</sub>                                      |
|                   |                                         | Fuel combustion                                                                | CO <sub>2</sub> , CH <sub>4</sub> , N <sub>2</sub> O |

Agricultural soil emissions (ASE) cover direct and indirect<sup>6</sup> emissions of nitrous oxide (N<sub>2</sub>O) from managed soils that are reported in the GHG inventories under the agriculture category following 2006 IPCC Guidelines. If emissions from managed agricultural soils in the activity are lower than in the baseline, those emissions shall be accounted in equation (2) under the term ASE. If emissions from managed agricultural soils in the activity are higher than in the baseline, those emissions shall be accounted in equations (1) or (2) as GHG<sub>associated</sub>.

Where net emissions from living biomass occur, they shall be accounted as GHG<sub>associated</sub>; the scope of these emissions shall only cover the biomass from trees and shrubs (stems, branches and roots).

Changes in soil carbon stock (CR in mineral soils), in mineral and organic soil emissions (LSE) and in N<sub>2</sub>O emissions (ASE) shall be quantified separately for each stratum of land and summed up across all strata of land, to obtain the total carbon stock change and changes in soil emissions in the activity area. In addition, changes in soil carbon stock or in soil emissions may be quantified for other areas under the operational control of the operator.

Where agriculture and agroforestry activities on mineral soils generate both carbon removals and soil emission reductions, the GHG<sub>associated</sub> shall be subtracted from both the temporary net carbon removal benefit and the soil emission reduction benefit after application of a weighting factor. The weighting factor shall correspond to the relative sizes of the gross temporary net carbon removal benefit and the gross soil emission reduction benefit, where 'gross' means before subtraction of the GHG<sub>associated</sub>.

Where CO<sub>2</sub> emissions from organic soils are reported, the estimate shall include off-site emissions such as Particulate Organic Carbon, Dissolved Organic Carbon, and Dissolved Inorganic Carbon.

<sup>6</sup> Indirect N<sub>2</sub>O emissions are emissions resulting from atmospheric deposition and nitrogen leaching and run-off.

### 2.1.1. *Agriculture and agroforestry on mineral soils*

The scope of the certification methodology for agriculture and agroforestry on mineral soils covers:

- (a) carbon removals in mineral soils and, in the event of agroforestry practices, in living biomass;
- (b) LULUCF soil emissions in mineral soils;
- (c) agriculture soil emissions from managed agricultural soils;
- (d) GHG<sub>associated</sub>, specifically:
  - (i) an increase in CO<sub>2</sub> emissions from living biomass, if applicable;
  - (ii) an increase in direct and indirect N<sub>2</sub>O emissions from agricultural soils resulting from changes in the application of fertilising products;
  - (iii) an increase in CO<sub>2</sub> emissions from liming or urea application;
  - (iv) an increase in fuel combustion emissions from field operations or transport to the activity area.

### 2.1.2. *Rewetting and restoration of peatlands and of other organic soils*

The scope of the certification methodology for rewetting and restoration of peatlands and of other organic soils covers:

- (a) LULUCF soil emissions in organic soils;
- (b) agriculture soil emissions from managed agricultural soils (optional);
- (c) GHG<sub>associated</sub>, specifically:
  - (i) an increase in CO<sub>2</sub> emissions from living biomass, if applicable;
  - (ii) an increase in direct and indirect N<sub>2</sub>O emissions from agricultural soils;
  - (iii) an increase in CO<sub>2</sub> emissions from liming or urea application, if applicable;
  - (iv) an increase in fuel combustion emissions from field operations or transport to the activity area.

### 2.1.3. *Afforestation*

The scope of the certification methodology for afforestation covers:

- (a) carbon removals in living biomass and, optionally, in mineral soils;
- (b) LULUCF soil emissions from mineral and organic soils (optional, except if tillage exceeds 15 cm of depth as referred to in Section 1.1.3.2, fourth paragraph);
- (c) agriculture soil emissions from managed agricultural soils (optional);
- (d) GHG<sub>associated</sub>, specifically:
  - (i) an increase in direct and indirect N<sub>2</sub>O emissions from agricultural soils;
  - (ii) an increase in fuel combustion emissions from field operations or transport to the activity area.

## 2.2. **Quantification formulas**

For the calculation of carbon removals, of LSE in mineral soils and, in the case of rewetting of peatlands and of other organic soils, of GHG<sub>associated</sub> from living biomass, the change in

carbon stock in the baseline scenario and in the activity scenario during each certification period shall be calculated in accordance with equation (3):

|                                                   |                 |
|---------------------------------------------------|-----------------|
| $change\ in\ carbon\ stock = C_{y=t+x} - C_{y=t}$ | equation<br>(3) |
|---------------------------------------------------|-----------------|

Where:

- y = year (y=0 is the year when the activity period starts)
- t = year in which a certification period starts
- x = length of the certification period (in years or months)
- C<sub>y=t</sub> = carbon stock at the start of the activity period (if t=0) or at the start of each certification period (if t ≥ 1) (in tonnes C)
- C<sub>y=t+x</sub> = carbon stock at the end of each certification period (in tonnes C)

The change in carbon stock shall be multiplied by –44/12 to convert carbon into CO<sub>2</sub>. Where the carbon stock increases, this results in carbon removals, which is a negative number. Where the carbon stock decreases, this results in LSE or GHG<sub>associated</sub>, which are positive numbers.

LSE in organic soils, ASE, and GHG<sub>associated</sub> (except those from living biomass), shall be calculated by adding together the yearly emissions of the relevant GHG in the baseline scenario and in the activity scenario during each certification period, in accordance with equation (4):

|                                      |                 |
|--------------------------------------|-----------------|
| $emissions = \sum_{y=t}^{y=t+x} E_y$ | equation<br>(4) |
|--------------------------------------|-----------------|

Where:

- y = year (y=0 is the year when the activity period starts)
- t = year in which a certification period starts
- x = length of the certification period (in years or months)
- E<sub>y</sub> = emissions (LSE, ASE, or GHG<sub>associated</sub>) in year y (in tonnes of CO<sub>2</sub> equivalent)

For the conversion of GHG emissions to CO<sub>2</sub> equivalent, the global warming potentials included in Commission Delegated Regulation (EU) 2020/1044<sup>7</sup> or in the latest IPCC Assessment Report<sup>8</sup> shall be used.

<sup>7</sup> Commission Delegated Regulation (EU) 2020/1044 of 8 May 2020 supplementing Regulation (EU) 2018/1999 of the European Parliament and of the Council with regard to values for global warming

## 2.3. Baselines

### 2.3.1. Baseline for soils

An activity-specific baseline shall apply to carbon removals in mineral soils, LSE and ASE.

The baseline scenario shall reflect the continuation of the soil or crop management carried out during a reference period considering, where applicable, crop rotation cycles. The reference period shall consist of at least the three calendar years immediately prior to the start of the activity period. The length of the reference period shall be extended, where applicable, to the length of the relevant crop rotation cycle.

Where the model approach described in Section 2.4.1 is used, the projected activity-specific baseline shall be included in the activity plan and, where appropriate, adjusted at re-certification audits to take into account the impacts of actual weather conditions that have occurred during the certification periods.

Where the measurement approach described in Section 2.4.2 is used, the baseline shall be measured in control plots where the same management as in the reference period is carried out, or a baseline equal to zero shall apply.

### 2.3.2. Baseline for living biomass

A baseline equal to zero shall apply to carbon removals in living biomass from newly planted trees, woody elements, or orchards. In case trees, woody elements, or orchards were present on the activity area prior to the time of submission of the activity plan to the certification body, the growth in carbon stock from such pre-existing vegetation shall not be included in the quantification of the temporary net carbon removal benefit.

### 2.3.3. Update of the baseline

For the practices referred to in Section 1.1.1.1, point (c), ASE<sub>baseline</sub> shall be updated every five years by applying a downward adjustment. Such adjustment shall correspond to 1 % of the initial baseline for every year that has passed since the start of the first activity period.

## 2.4. Quantification and monitoring approaches

One or more of the following approaches shall be used to quantify and monitor the temporary net carbon removal benefit and the net soil emission reduction benefit:

- (a) approach 1: models
- (b) approach 2: measurements
- (c) approach 3: default emission factors

Table 4 lists the possible approaches to quantify and monitor each term of the temporary net carbon removal benefit and the net soil emission reduction benefit:

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potentials and the inventory guidelines and with regard to the Union inventory system and repealing Commission Delegated Regulation (EU) No 666/2014 (OJ L 230, 17.7.2020, p. 1, ELI: [http://data.europa.eu/eli/reg\\_del/2020/1044/oj](http://data.europa.eu/eli/reg_del/2020/1044/oj)).

<sup>8</sup> Sixth Assessment Report (AR6) of the IPCC 2021 Forster, P., T. Storelvmo, K. Armour, W. Collins, J. L. Dufresne, D. Frame, D. J. Lunt, T. Mauritsen, M. D. Palmer, M. Watanabe, M. Wild, H. Zhang, 2021, The Earth's Energy Budget, Climate Feedbacks, and Climate Sensitivity. In: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press. In Press. - [https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC\\_AR6\\_WGI\\_Chapter\\_07.pdf](https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Chapter_07.pdf)

Table 4

| IPCC category     | Term                            | IPCC carbon pool or GHG emission source                                        | Approach(es) |
|-------------------|---------------------------------|--------------------------------------------------------------------------------|--------------|
| <b>LULUCF</b>     | <b>CR</b>                       | Living biomass                                                                 | 1, 2         |
|                   |                                 | Soil Organic Carbon in mineral soils                                           | 1, 2         |
|                   | <b>LSE</b>                      | Soil Organic Carbon in mineral soils                                           | 1, 2         |
|                   |                                 | Soil Organic Carbon in organic soils                                           | 1            |
| <b>Non-LULUCF</b> | <b>ASE</b>                      | Direct and indirect N <sub>2</sub> O emissions from managed agricultural soils | 1, 3         |
|                   | <b>GHG<sub>associated</sub></b> | Living biomass                                                                 | 1, 2, 3      |
|                   |                                 | Direct and indirect N <sub>2</sub> O emissions from managed agricultural soils | 1, 3         |
|                   |                                 | Liming & urea application                                                      | 1, 3         |
|                   |                                 | Fuel combustion                                                                | 3            |

Where multiple approaches are applicable, only one of them shall be chosen to quantify or monitor the same term in a consistent way throughout the monitoring period.

The same approach shall be used for the calculation of the same term in the activity and in the baseline scenarios.

#### 2.4.1. Approach 1: models

##### 2.4.1.1. Eligible models

Eligible models shall cover:

- (a) process-based models, that is, computational frameworks that use scientific theories and mechanistic equations to simulate biological, chemical, and physical processes governing carbon and/or nitrogen dynamics, including biogeochemical models, soil carbon models, dynamic vegetation models, ecosystem process models, and hydrological models;
- (b) data-driven models, that is, analytical methodologies that derive predictions through empirical analysis, and that employ statistical techniques to evaluate observational data sets related to carbon stocks and GHG fluxes, relying primarily on evidence extracted from data trends to infer changes in carbon stocks and GHG fluxes due, inter alia, to management practices; examples include machine learning models, statistical regression models (such as correlative GHG flux proxy approaches), geospatial analysis models (such as digital soil mapping tools), remote sensing-based models (such as models using biomass-based data from LiDAR or high-resolution satellite data), and Bayesian models.

The selected model shall be either a model used for the compilation of the national GHG inventory of the Member State where the activity takes place or a model that the Commission considers compliant with the following criteria:

- (a) transparency and traceability, that is to say, the attributes of the model shall be well described, the model version shall be clearly indicated and changes between versions shall be traceable and well documented;
- (b) scientific credibility, that is to say, the model shall have been shown in peer-reviewed scientific studies to estimate carbon stock changes and/or GHG emissions for soil/crop management, fertiliser management, rewetting and restoration of peatlands and of other organic soils, or agroforestry/afforestation under comparable pedo-climatic conditions and for the plant functional types or vegetation characteristics that are relevant to the activity;
- (c) suitability, that is to say, the model shall have been calibrated and validated based on direct measurements from datasets that accurately represent the pedo-climatic region where the activity takes place, while the calibration and validation data shall be independent, either by using two separate datasets with no overlap of research locations, or by employing a statistical process such as cross-validation;
- (d) minimum accuracy, that is to say, when tested against the validation dataset, the mean bias of the model, as expressed in equation (5), shall be smaller than or equal to an estimate of pooled measurement uncertainty (PMU), as expressed in equation (6); at least 90 % of the observed values shall fall within the model's 90 % prediction intervals, and the amount of variance explained by the model ( $R^2$ ), as expressed in equation (7), shall be greater than zero.

|                                                 |                 |
|-------------------------------------------------|-----------------|
| $\text{mean bias} = \sum_{i=1}^n (P_i - O_i)/n$ | equation<br>(5) |
|-------------------------------------------------|-----------------|

Where:

$P_i$  = predicted change in carbon stock or GHG emission

$O_i$  = observed change in carbon stock or GHG emission

$i$  = index of observation within study

$n$  = number of observations for a given study

|                                                                                 |                 |
|---------------------------------------------------------------------------------|-----------------|
| $PMU = \sqrt{\frac{\sum_{j=1}^k \sigma_j^2 (n_j - 1)}{\sum_{j=1}^k (n_j - 1)}}$ | equation<br>(6) |
|---------------------------------------------------------------------------------|-----------------|

Where:

$k$  = number of observations across all studies

$\sigma_j$  = standard error of the  $j^{\text{th}}$  observed change in carbon stock or GHG emission

$n_j$  = number of replicate measurements used in the  $j^{\text{th}}$  observation

|                                                                               |                 |
|-------------------------------------------------------------------------------|-----------------|
| $R^2 = 1 - \frac{\sum_{i=1}^n (P_i - O_i)^2}{\sum_{i=1}^n (O_i - \bar{O})^2}$ | equation<br>(7) |
|-------------------------------------------------------------------------------|-----------------|

Where:

$P_i$  = predicted change in carbon stock or GHG emission

$O_i$  = observed change in carbon stock or GHG emission for observation i

$\bar{O}$  = mean of observed values of change in carbon stock or GHG emission

$i$  = index of observation within study

$n$  = number of observations in the validation set

#### 2.4.1.2. Use of models

Where a model is used to estimate changes in Soil Organic Carbon (SOC), it shall be initialised with the SOC stock that shall be determined based on soil samples taken at the start of the activity period and optional complementary data, such as soil maps. To this end, the locations of soil samples shall be determined in accordance with the rules on the use of measurements referred to in Section 2.4.2.2, and a representative selection of at least 20 % of those locations, covering the main strata, shall be sampled.

At least every five years, those sampling points shall be resampled at the same locations for verification of the model results, using the same sampling and lab analysis approach. The results shall be taken into account in the uncertainty quantification of the model used in future certification periods.

Alternatively, any available data from regional or national monitoring networks with similar density of sampling points may be used for model verification.

Where a model is used to estimate GHG fluxes of organic soils using vegetation, water table levels and land use as proxies, the following requirements shall apply:

- (a) land use may only be used as the main proxy if water table level measurements and vegetation data are not available, e.g. in the case of bare peat;
- (b) the GHG flux model shall be based on a meta-analysis of published data and be calibrated and validated with on-site measurements in the same pedo-climatic region as the activity area and with similar vegetation type;
- (c) relevant vegetation types shall be distinguished on the basis of plant species or species groups that are indicative for water table depth and on the proportion of shunt species in the vegetation;
- (d) if a vegetation type in the activity area does not have sufficient similarity with the vegetation types for which GHG fluxes have been derived, regression models between the mean annual water table depth and GHG fluxes may be used to determine the emission factors;

- (e) the activity area shall be mapped at least at 1:5000 scale with the vegetation being documented every five years and the water table level relative to the soil surface being measured or modelled at least every 15 days.

Where a model is used to estimate changes in carbon stock in living biomass based on remote sensing data, the model shall take into account the most prominent site-specific variables pertinent to the activity, including land cover, volume of above-ground biomass, volume of below-ground biomass, local climate conditions, soil type, management type and intensity, and digital elevation model. The minimum resolution of land cover and volume of above-ground biomass shall be 10 m. The processing of raw data from remote sensing shall include local calibration supported by the use of measurements determined and incorporated in the final calculation of the confidence intervals for the estimates of total carbon removals or soils emission reductions.

#### 2.4.1.3. Information to be included in the monitoring plan

The monitoring plan shall include a description of the procedures that were followed to apply the model, covering at least:

- (a) conformity of the use of the model with the validated model domain;
- (b) a description of the type and source of input data and potential pre-processing of the data;
- (c) the system boundaries (spatial, temporal, soil depth, etc.);
- (d) a description of the approach taken to estimate uncertainty in the model outputs;
- (e) information on the quality assurance and quality control steps;
- (f) the responsibility for data collection and archiving.

#### 2.4.2. *Approach 2: measurements*

##### 2.4.2.1. Eligible measurements

Eligible measurements shall cover:

- (a) direct measurements of SOC through soil sampling techniques;
- (b) allometric equations and biomass expansion factors (BEFs) that indirectly measure biomass by combining direct measurements of forest structure, such as tree diameter and tree height, with correlative relationships;
- (c) indirect measurements through on-site proximal sensing techniques, that is, field-based sensors that are in contact with, or within 2 m of the soil and that detect signals from the soil.

Allometric equations or BEFs referred to in the first paragraph, point (b), shall be one of the following:

- (a) site-specific, that is, developed by harvesting a representative sample of similar trees on a site close to the activity area and directly measuring their volumes and dry biomass;
- (b) published in peer-reviewed scientific studies carried out in the location or region of the activity or for similar forest type and climate and edaphic conditions;
- (c) applied in regional or national forest inventories, or in the national GHG inventories for LULUCF.

In addition, allometric equations and BEFs shall be selected in accordance with the following criteria:

- (a) taxonomic hierarchy validation: the choice of the equation shall fit the species being measured and minimise bias introduced by phylogenetic differences in tree architecture and wood properties, and follow the following priority order:
  - (i) species-specific equations developed precisely for the planted or inventoried species;
  - (ii) genera-specific equations;
  - (iii) equations derived from closely related families or functional group;
  - (iv) broadleaved or coniferous generalised equations, subject to justification of applicability across a wide functional group;
- (b) size and stand structure validation: the distribution of tree sizes (stand structure) in the inventory plot shall align with the range utilised for the equation's original development;
- (c) climatic and edaphic validation: the climatic and edaphic conditions of the activity area shall fall within the range of conditions under which the selected equation was originally developed.

Proximal sensing techniques referred to in the first paragraph, point (c), shall be techniques that the Commission has considered compliant with the following criteria:

- (a) transparency, that is to say, the functions and parameters underlying the proximal sensing technique shall be well described;
- (b) scientific credibility, that is to say, the proximal sensing technique shall have been shown in peer-reviewed scientific studies to be sensitive enough to detect variations in soil or biomass carbon stocks across different pedo-climatic conditions or management practices, and, if used to measure soil carbon, to specifically measure soil carbon without significant interference from other soil properties;
- (c) suitability, as described in Section 2.4.1.1., second paragraph, point (c);
- (d) minimum accuracy, as described in Section 2.4.1.1., second paragraph, point (d).

#### 2.4.2.2. Use of measurements

*A. Rules applying to the approaches referred to in Section 2.4.2.1., first paragraph, points (a) and (c)*

Sampling design for mineral soils may be based on representative stratified sampling of the activity area consisting of design, model (grid, geo-statistical) or a combination (hybrid) sampling design. Sampling design and strata shall remain the same between measurement and remeasurement. Sample size shall be calculated ex ante using confidence interval or hypothesis testing methods and be large enough to quantify changes at the lower bound of the one-tailed confidence interval with minimum confidence level of 70 %. The minimum number of samples shall be calculated in accordance with equation (8):

|                                            |                 |
|--------------------------------------------|-----------------|
| $n \geq \sigma^2 \frac{z_{\alpha}^2}{E^2}$ | equation<br>(8) |
|--------------------------------------------|-----------------|

Where:

- $n$  = expected minimum number of samples
- $\sigma^2$  = expected variance of SOC based on existing local datasets
- $z_\alpha$  = critical value of the standard normal distribution at the significance level  $\alpha$  ( $\alpha$  = 100 % - confidence level), with a z-value of 1,04 for a 70 % confidence level
- $E$  = tolerable error (e.g. expected effect size multiplied by the confidence level or the minimum detectable difference)

Alternatively, the methodology set out in Annex V to Commission Implementing Regulation (EU) 2022/996<sup>9</sup> may be applied.

Field sampling locations shall be determined from the sampling design and sample size. The coordinates of the sampling locations shall be recorded to support resampling at the same locations.

Individual or composite sampling of soils at sampling locations may be carried out, in accordance with the required sample size and sampling design.

Soil shall be sampled to a minimum depth of 30 cm or, in the case of shallower topsoil, to the bedrock. Actual soil depths sampled shall be recorded and used in SOC stock calculations.

Soil sampling shall be carried out within the same season, at least six weeks after the last organic fertilisation event and, where possible, under the same soil and tillage conditions at measurement and remeasurement.

The same equipment shall be used for measurement and remeasurement.

*B. Rules applying to the approach referred to in Section 2.4.2.1, first paragraph, point (a)*

Laboratory determination of SOC stocks shall be based on equation (9):

|                                                                                                                |                 |
|----------------------------------------------------------------------------------------------------------------|-----------------|
| $\text{SOC}_{\text{stock}} = \text{OC}_{\%} \times \text{BD}_{\text{fe}} \times \text{depth} \times (1 - G_v)$ | equation<br>(9) |
|----------------------------------------------------------------------------------------------------------------|-----------------|

where:

- $\text{OC}_{\%}$  = organic carbon content in the fine earth fraction (% , < 2 mm)
- $\text{BD}_{\text{fe}}$  = bulk density of the fine earth fraction ( $\text{g cm}^{-3}$ )
- depth = depth of the respective soil layer (cm) from field sampling
- $G_v$  = volumetric coarse fragment (> 2 mm) content (a default density value of 2,6 g/cm<sup>3</sup> may be used to convert mass into volume)

Organic Carbon content may be determined using dry combustion with elemental analysis (ISO 10694:1995), near-infrared spectrometry (ISO 17184:2014) or alternative methods for which an ISO standard or a formal accreditation in a Member State is available. Inorganic carbon shall be excluded.

<sup>9</sup> Commission Implementing Regulation (EU) 2022/996 of 14 June 2022 on rules to verify sustainability and greenhouse gas emissions saving criteria and low indirect land-use change-risk criteria (OJ L 168, 27.6.2022, p. 1, ELI: [http://data.europa.eu/eli/reg\\_impl/2022/996/oj](http://data.europa.eu/eli/reg_impl/2022/996/oj)).

BD<sub>fe</sub> shall be determined using either a laboratory determination following ISO 11272:2017, with the addition of documented steps to determine BD<sub>fe</sub> from the fine earth fraction; or alternative methods for which ISO standards or a formal accreditation in a Member State are available; or a validated pedo-transfer function to predict BD<sub>fe</sub>. Such a pedo-transfer function shall be representative of the pedo-climatic conditions, based on data from at least 200 direct measurements and have an R<sup>2</sup> of at least 0,6.

Where relevant, models and model parameters shall be clearly described and justified. For all methods, analytical error shall be reported.

For remeasurement, the same BD<sub>fe</sub> and G<sub>v</sub> values used for the initial SOC stock may be used to calculate the SOC stock at the subsequent re-certification audits. Where the BD<sub>fe</sub> is measured again, the SOC stock changes shall be calculated based on the equivalent soil mass (ESM) method as described in the UN Food and Agriculture Organization Guidelines (2019)<sup>10</sup> or other ESM methods, where justified. Errors associated with ESM methods shall be accounted for with error propagation. ESM methods and calculations shall be documented. ESM is not required where soils are sampled to a depth of at least 60 cm or to bedrock.

Laboratory methods shall remain the same and demonstrate consistency for measurement and remeasurement. Laboratories used for soil analysis shall be in conformity with ISO/IEC 17025:2017 and make available quality control data (e.g. interlaboratory trials, use of reference materials, calibration methods) for the determination of analytical uncertainty and audit purposes.

*C. Rules applying to the approach referred to in Section 2.4.2.1, first paragraph, point (b)*

Sampling design for agroforestry shall follow a census-based approach: at the start of the activity period, a count of planting units (single trees, lines of trees, meter of hedges, small copses) shall be performed. The number of planting units shall be used for scaling of carbon stocks. At subsequent re-certification audits, an appropriate representative sample of the planting units shall be used. Biomass of trees and hedges shall be estimated for each parcel and for parcel boundaries. Locations for sampled trees or hedges shall be distributed randomly across the activity area.

Sampling design for afforestation shall follow an area-based approach: plot-based sampling methods shall be used to scale biomass and resulting carbon stock estimates per unit area, using the activity area as the multiplier. Sample plots shall provide a representative sample and be distributed randomly in all strata identified across the activity area. The number of plots shall be selected depending on the variability of the activity area and the accepted margin of error.

For agroforestry and afforestation, field sampling locations shall be determined from the sampling design and sample size. The coordinates of the sampling locations shall be recorded to support resampling at the same locations. Tree sizes like diameter and height shall be measured and converted to biomass and carbon using appropriate allometric equations and/or BEFs as referred to in Section 2.4.2.1 Information on species and size of individual trees shall be recorded together with the applied allometric equations or BEFs.

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<sup>10</sup> Food and Agriculture Organization of the United Nations, Measuring and modelling soil carbon stocks and stock changes in livestock production systems, Guidelines for assessment, 2019.

For agroforestry, the number of planting units shall be selected depending on species and diameter classes of the trees, the biophysical circumstances in the activity area, the desired confidence interval and the accepted margin of error. The minimum number of planting units to be sampled shall be calculated using equation (10):

|                                            |                  |
|--------------------------------------------|------------------|
| $n \geq \sigma^2 \frac{z_{\alpha}^2}{E^2}$ | equation<br>(10) |
|--------------------------------------------|------------------|

Where:

- $n$  =        expected minimum number of sampling units
- $\sigma^2$  =        expected variance of biomass based on existing local datasets or new measurements
- $z_{\alpha}$  =        critical value of the standard normal distribution at the desired significance level  $\alpha$  ( $\alpha$  = 100 % - confidence level), with a z-value of 1,65 for a 90 % confidence level.
- $E$ =        tolerable error (e.g. expected effect size multiplied by the confidence level)

For afforestation, the number of plots that shall be sampled per stratum may be calculated using the relevant clean development mechanism afforestation and reforestation (CDM A/R) Methodological Tool<sup>11</sup> or available tools adopted under the Paris Agreement Crediting Mechanism.

2.4.2.3. Information to be included in the monitoring plan

Where the approaches referred to in Section 2.4.2.1, first paragraph, points (a) or (c), are used, the monitoring plan shall include:

- (a)        a description of the sampling design, clearly describing, if applicable, methods and models used, including model parameters;
- (b)        a demonstration that the sample size is appropriate for the sampling design deployed and sufficient to measure expected carbon stock changes for the activity area over the certification period, based on relevant scientific literature;
- (c)        the minimum relevant effect/difference, that is, the expected carbon stock change over the certification period that shall be detected by the sampling design;
- (d)        prior information on the population variance of the target variable, that is, the expected variability in carbon stock changes for the activity area;
- (e)        a description and justification of the process for selecting sample sites, field procedures to collect measurements, measurement equipment and methods, and processes for dealing with exclusions and replacements;
- (f)        a sample collection and relevant quality assurance and quality control protocols;
- (g)        a description of the re-sampling strategy and reasoning for the adopted resampling frequency;

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<sup>11</sup> UNFCCC/CCNUCC A/R Methodological Tool Calculation of the number of sample plots for measurements within A/R CDM project activities (Version 02.1.0).

- (h) data processing and quality assurance and quality control protocols including how adopted re-sampling period is handled when making carbon stock change estimates and their associated uncertainty.

#### 2.4.3. *Approach 3: default emission factors*

##### 2.4.3.1. Eligible emission factors

Eligible emission factors shall cover:

- (a) Tier 2 emission factors used in the national GHG inventory of the Member State where the activity takes place;
- (b) where emission factors under point (a) are not available, Tier 1 emission factors included in the 2006 IPCC Guidelines<sup>12</sup> and any further refinement to those Guidelines.

##### 2.4.3.2. Use of emission factors

Emission factors shall be multiplied by the relevant activity data.

##### 2.4.3.3. Information to be included in the monitoring plan

Where approach 3 is used, the monitoring plan shall include:

- (a) a justification of the selected emission factors;
- (b) the source of the used emission factors;
- (c) a description of the activity data.

#### 2.5. **Quantification of the uncertainty deduction factor**

The uncertainty deduction factor shall be 10 % or equal to the estimated uncertainty, whichever is higher. To this end, the quantification of total carbon removals and soil emissions shall include an estimate of uncertainty by considering the model prediction error, the measurement error of input data, and the sampling error related to the sampling design, as applicable.

The model prediction error shall be quantified using either statistical validation based on ground-truth measurements or a Monte Carlo simulation. Where a frequentist statistical validation method is used, the model prediction error shall be expressed as the standard deviation of the difference between the modelled values and ground-truth measurements; when a Monte Carlo simulation is used, the model prediction error shall be expressed as the standard deviation of the probability distribution. The input data for Monte Carlo simulation shall be drawn from distributions described by the mean inputs of the measured parameters and their corresponding measurement error.

The measurement error shall correspond to the standard error of the mean and shall be obtained from:

- (a) laboratories used for soil analysis, for mineral soils;
- (b) default values in the 2006 IPCC Guidelines, expert judgment, or statistical sampling methods, for organic soils;
- (c) the 2006 IPCC Guidelines, for biomass;

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<sup>12</sup> [https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4\\_Volume4/V4\\_11\\_Ch11\\_N2O&CO2.pdf](https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_11_Ch11_N2O&CO2.pdf)

(d) information from producers of measurement equipment.

For biomass, the sampling error shall correspond to the standard error of the mean and be calculated based on repeated sampling of some of the fields. For mineral soils, the sampling error shall correspond to the standard error of the mean and be calculated based on repeated sampling of some of the fields or on local information on SOC variability from existing sources. When using BEFs, uncertainty shall be propagated in the resulting carbon stock changes for living biomass in line with the 2006 IPCC Guidelines on combining uncertainties<sup>13</sup>. For organic soils, estimates of the sources of uncertainty may be used if they are based on credible literature or expert judgment.

The uncertainty deduction factor shall be computed in accordance with equation (11), where models are used, and in accordance with equation (12), where direct measurements are used:

|                                                                     |               |
|---------------------------------------------------------------------|---------------|
| $UNC = \left( \frac{SD_{\Delta x}}{\Delta x} \right) \times t_{ci}$ | equation (11) |
| $UNC = \left( \frac{SE_{\Delta x}}{\Delta x} \right) \times t_{ci}$ | equation (12) |

Where:

|                 |                                                                                                                                                                            |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UNC             | = uncertainty deduction factor (fraction between 0,1-1)                                                                                                                    |
| $SD_{\Delta x}$ | = standard deviation of the mean modelled carbon stock change of carbon pool x                                                                                             |
| $SE_{\Delta x}$ | = standard error of the mean measured carbon stock change of carbon pool x                                                                                                 |
| $\Delta x$      | = mean estimated stock change of carbon pool x                                                                                                                             |
| $t_{ci}$        | = t value for a one-sided student's t-distribution at confidence interval ci, which shall be equal to 70 % for SOC stock changes and 90 % for biomass carbon stock changes |

For N<sub>2</sub>O emissions from managed soils, the uncertainty deduction factor shall be determined in accordance with equation (13).

|                                                                                 |               |
|---------------------------------------------------------------------------------|---------------|
| $UNC_{ASE} = \left( \frac{SD_{\Delta N_2O}}{\Delta N_2O} \right) \times t_{ci}$ | equation (13) |
|---------------------------------------------------------------------------------|---------------|

Where:

|                    |                                                                                    |
|--------------------|------------------------------------------------------------------------------------|
| $UNC_{ASE}$        | = uncertainty deduction factor for ASE (fraction between 0,1-1)                    |
| $SD_{\Delta N_2O}$ | = standard deviation of the mean modelled N <sub>2</sub> O soil emission reduction |
| $\Delta N_2O$      | = mean estimated N <sub>2</sub> O soil emission reduction                          |

<sup>13</sup> [https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/1\\_Volume1/V1\\_3\\_Ch3\\_Uncertainties.pdf](https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/1_Volume1/V1_3_Ch3_Uncertainties.pdf) and [https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/1\\_Volume1/19R\\_V1\\_Ch03\\_Uncertainties.pdf](https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/1_Volume1/19R_V1_Ch03_Uncertainties.pdf)

$t_{ci}$  = t value for a one-sided student's t-distribution at confidence interval  $ci$  which shall be equal to 70 %

When emission factors are used to quantify and monitor N<sub>2</sub>O emissions from managed soils, the uncertainty deduction factor shall be 10 %.

### **3. ADDITIONALITY**

#### **3.1. Regulatory test**

This test shall demonstrate that the net climate benefit resulting from the activity does not occur as a result of any legal requirement on the operator stemming from Union or national legislation to carry out the activity in the activity area, unless such legislation refers to or formally integrates certification under Regulation (EU) 2024/3012 as an instrument for implementation.

National legislation includes laws, statutes and regulations also at regional level, as well as court orders, mandatory environmental management agreements or planning decisions or other legally binding agreements.

Activities going beyond the level required under national or Union obligations may generate units eligible for certification for those additional emission reductions or carbon removals.

#### **3.2. Financial viability**

##### *3.2.1. Incentive effect test*

This test shall demonstrate that certification has an incentive effect, that is that certification induces the operator to change its behaviour in order to engage in the additional activity, which it would not carry out without certification or would carry out in a restricted or different manner.

Certification presents an incentive for the operator wherever work on the relevant activity has not yet started at the time of submission of the application to the certification scheme. In this regard, start of works on the activity means the earlier of, either the start of the activity, or the first legally binding commitment to order equipment or employ services or any other commitment that makes the activity irreversible. Preparatory works such as obtaining permits and conducting feasibility studies are not considered start of works or activity.

By way of derogation from the first and second paragraphs, the obligation to carry out the incentive effect test does not apply to:

- (a) operators who started the activity before the submission of the activity plan to the certification scheme, if the activity started between 1 January 2023 and 31 December 2027. In this case, for the quantification of the baseline the reference period referred to in Section 2.3.1 shall consist of at least the three years immediately prior to the start of works on the relevant activity, and the quantification approach referred to in Section 2.3.2 may include trees and vegetation that were present on the activity area prior to the submission of the application to the certification scheme, if they were planted between 1 January 2023 and 31 December 2027;

- (b) operators who started the activity under a certification scheme before its recognition under Implementing Regulation (EU) 2025/2358, if the activity started between 1 January 2023 and 31 December 2030.

### 3.2.2. *Financial viability tests*

The financial viability tests shall demonstrate that the activity is not financially viable in the absence of revenues from certification by means of an investment analysis.

To that end, one of the following tests may be used:

- (a) simple cost test;
- (b) investment comparison test.

#### 3.2.2.1. *Simple cost test*

The simple cost test shall demonstrate that the implementation of the activity is associated with costs and does not generate any cost savings or revenues other than revenues from certification during the monitoring period.

The simple cost test shall be conducted using the following steps:

- (a) describing the costs associated with the financing of the activity over the monitoring period;
- (b) demonstrating with appropriate evidence that, in the absence of certification, no cost savings or revenues would be generated during the monitoring period;
- (c) documenting any public funding provided to the activity and demonstrating that public funding would not have filled the funding gap of the activity in the absence of the revenues from certification, considering capital expenditure (CAPEX) and operational expenditure (OPEX) and any revenues other than revenues generated from certification.

#### 3.2.2.2. *Investment comparison test*

The investment comparison test shall compare the financial attractiveness of the activity with alternative investment scenarios.

The investment comparison test shall be conducted using the following steps:

- (a) describing the costs associated with the financing of the activity and of the alternative scenarios to be considered over the monitoring period;
- (b) using the net present value (NPV) of the investment as indicator for assessing the financial viability of the activity and of the alternative scenarios;
- (c) calculating the financial viability of the activity and of the alternative scenarios without revenues from the certification, using the NPV as calculated in accordance with equation (14), justifying the calculation with appropriate evidence;

|                                      |                  |
|--------------------------------------|------------------|
| $NPV = \sum \frac{P - L}{(1 + i)^t}$ | equation<br>(14) |
|--------------------------------------|------------------|

Where:

- P = expected revenues from the activity (without revenues from certification)
- L = expected costs of the activity

- i = estimated weighted average cost of capital or discount rate of 3,5 %<sup>14</sup>
- t = monitoring period;
- (d) applying Section 3.2.2.1, second paragraph, point (c);
- (e) demonstrating that the activity without revenues from certification is not the most financially viable scenario, by comparing the NPV calculated for the activity and the alternative scenarios.

### 3.2.2.3. Common rules to simple cost and investment comparison tests

The tests shall include all relevant costs, including CAPEX and OPEX, and all revenues and cost savings, including any public funding received. The tests may also include the costs associated with any barriers faced by the proposed activity where they can be monetised and quantified as additional costs.

The tests shall not include expenditures incurred prior to the start of the activity, save where such expenditures were committed prior to the start date, but payment is to be made after the start date.

All parameters and assumptions shall be internally consistent. The assumptions, data and conclusions shall be transparently documented and appropriately justified and substantiated by evidence.

The test shall be implemented in a conservative manner.

## 4. STORAGE, MONITORING AND LIABILITY

### 4.1. Risk assessment

Operators or groups of operators shall carry out a risk assessment for the activity before the start of the activity period. The risk rate resulting from the risk assessment of the activity shall account only for unavoidable events.

Reversals resulting from the management of the activity and any other wilful actions by the operator or from mismanagement, neglect or illegal actions by the operator, including as a consequence of insolvency, or reversals that have occurred following a failure to implement the risk mitigation measures that is attributable to the operator, shall be classified as avoidable.

Reversals resulting from natural events and extreme events, war or acts of terrorism, changes in policies or legal requirements that prevent the operators from implementing risk mitigation measures and reversals resulting from illegal action by third parties that may not be controlled, influenced or managed using legal means by the operator shall be classified as unavoidable.

Reversals caused by any factors that were not identified in the risk assessment shall be classified as avoidable and may only be classified as unavoidable with due justification, such as in cases of clear *force majeure*.

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<sup>14</sup> Based on Commission Implementing Regulation (EU) 2022/996 of 14 June 2022 on rules to verify sustainability and greenhouse gas emissions saving criteria and low indirect land-use change-risk criteria (OJ L 168, 27.6.2022, p. 11, ELI: [http://data.europa.eu/eli/reg\\_impl/2022/996/oj](http://data.europa.eu/eli/reg_impl/2022/996/oj)).

If, within the monitoring period, the total emissions from the release of stored carbon do not exceed the total expected carbon removals, such emissions shall not qualify as reversal.

Operators or groups of operators shall update the risk assessment every five years and after any unavoidable event.

#### 4.1.1. Agriculture on mineral soils

Operators or groups of operators shall calculate a risk rate for the activity, that shall be based on two indicators<sup>15</sup>:

- (a) hazard, that is, the climatic conditions and land management that drive changes in SOC, and for which recent changes in SOC in the activity area shall be used as a proxy;
- (b) vulnerability, that is, the characteristics of the soil that make it more prone to losing carbon in the case of hazard, and for which the level of carbon saturation in the activity area shall be used as a proxy.

For the purposes of the first paragraph, point (a), the changes in SOC in the activity area during the 10 years preceding the start of the activity period shall be estimated. To that end, operators or groups of operators may use own data, national survey data or the most recent estimate of SOC changes based on data made available by the Commission<sup>16</sup>.

The hazard indicator shall be classified as follows:

- (a) low, where SOC has increased or remained stable;
- (b) high, where SOC has decreased.

For the purposes of the first paragraph, point (b), carbon saturation shall be calculated as the ratio between the average mineral-associated organic carbon (MAOC) content in the activity area and the maximum MAOC content in the pedo-environmental zone to which the activity area belongs, expressed as a percentage.

To that end, carbon saturation values made available by the Commission<sup>17</sup> may be used. Alternatively, operators or groups of operators may measure MAOC levels in the activity area as the organic carbon associated to the fine fraction of the soil (that is, silt and clay) that can be isolated by a size fractionation (<53micro meter) of the bulk soil<sup>18</sup>, and divide it by the maximum MAOC content in the relevant pedo-environmental zone, as made available by the Commission<sup>19</sup>.

The vulnerability indicator shall be classified as follows:

- (a) low, where the carbon saturation is smaller than 68 %;

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<sup>15</sup> Breure, T.S., De Rosa, D., Panagos, P., Cotrufo, M.F., Jones, A. and Lugato, E., 2025. Revisiting the soil carbon saturation concept to inform a risk index in European agricultural soils. *Nature Communications*, 16(1), p.2538. <https://doi.org/10.1038/s41467-025-57355-y>

<sup>16</sup> <https://esdac.jrc.ec.europa.eu/esdacviewer/euso-dashboard/mini-dashboard/>

<sup>17</sup> Ibidem

<sup>18</sup> Lugato, E., Lavalley, J.M., Haddix, M.L. *et al.* Different climate sensitivity of particulate and mineral-associated soil organic matter. *Nat. Geosci.* 14, 295–300 (2021). <https://doi.org/10.1038/s41561-021-00744-x>

<sup>19</sup> <https://esdac.jrc.ec.europa.eu/esdacviewer/euso-dashboard/mini-dashboard/>

- (b) high, where the carbon saturation is equal to or larger than 68 %.

Based on the combination of the hazard and vulnerability indicators, the risk rate (expressed as the share of carbon at risk of being released) shall be determined in accordance with Table 50.

Table 5

|               |      | Hazard |      |
|---------------|------|--------|------|
|               |      | Low    | High |
| Vulnerability | Low  | 2 %    | 6 %  |
|               | High | 4 %    | 10 % |

Groups of operators may use the weighted average of the risk rate across the activity area.

Where risk mitigation practices beyond the eligibility and minimum sustainability requirements are put in place, such as use of deep-rooted crops and diversified crop sequences, the risk rate may be reduced by multiplying it by a factor ranging between 0,8 and 1. Certification schemes may provide additional guidance on these risk mitigation practices and the corresponding factor.

#### 4.1.2. Rewetting and restoration of peatlands and of other organic soils

Operators or groups of operators shall determine a risk level for the activity, that shall be based on two indicators:

- (a) disturbance regime in the activity area;
- (b) water exploitation.

For the purposes of the first paragraph, point (a), the return period of natural disturbance events in the activity area shall be calculated as the average interval between occurrences of a natural disturbance event. Natural disturbance events shall include at least forest fires.

To that end, the datasets on the prediction of the return period, made available by the Commission<sup>20</sup>, may be used. Alternatively, statistics on natural disturbance events covering a period of at least 30 years preceding the start of the activity period may be used to calculate the ratio between the number of natural disturbance events and the given time-period. In the latter case, such ratio shall be estimated using one or more of relevant empirical records, properly documented local surveys, Earth observation datasets, and peer reviewed literature. The area used to calculate return period statistics shall be large enough to allow for the quantification of robust return period statistics for disturbance regimes representative of the geographical region, ecological conditions or administrative level (NUTS2 or NUTS1).

The disturbance regime indicator shall be classified as follows:

- (1) low, where the return period is higher than or equal to 30 years;
- (2) medium, where the return period is between 15 and 30 years;
- (3) high, where the return period is below 15 years.

For the purposes of the first paragraph, point (b), the available average of the Water Exploitation Index plus (WEI+)<sup>21</sup> shall be used.

The water exploitation indicator shall be classified as follows:

<sup>20</sup> <https://data.jrc.ec.europa.eu/collection/id-00445>

<sup>21</sup> <https://data.europa.eu/data/datasets/byobiw86oozdduikcwmlxg?locale=en>

- (1) low, where the average WEI+ is less than 20 %;
- (2) medium, where the average WEI+ is between 20 % and 40 %;
- (3) high, where the average WEI+ is higher than 40 %.

Based on the combination of the disturbance regime and water exploitation indicators, the risk level shall be determined in accordance with Table 6.

Table 6

|                    |        | Water exploitation |             |             |
|--------------------|--------|--------------------|-------------|-------------|
|                    |        | Low                | Medium      | High        |
| Disturbance regime | Low    | Low risk           | Low risk    | Medium risk |
|                    | Medium | Low risk           | Medium risk | Medium risk |
|                    | High   | Medium risk        | Medium risk | High risk   |

In the case of medium-risk activities, operators shall put in place risk mitigation practices beyond the eligibility and minimum sustainability requirements, such as rising humidity levels by increasing the vegetation cover to reduce the risk of fires, or the reduction of water extraction in the river basin to make a sustainable use of the freshwater resources. Certification schemes may provide additional guidance on these risk mitigation practices.

#### 4.1.3. Afforestation and agroforestry

Operators or groups of operators shall assess the suitability of the species and species composition at the beginning of the activity period and at the end of the monitoring period. A tree species shall be considered suitable where its climate suitability, based on the species or its provenance, is determined to stay within modelled optimal climate niche for that species.

To that end, the dataset on current and future species suitability, made available by the Commission<sup>22</sup>, may be used. Alternatively, operators or groups of operators may use dedicated national assessments or scientifically validated climate envelope models published in peer-reviewed literature and based on climate scenarios modelled under Representative Concentration Pathway 4.5 or more conservative climate scenarios.

Operators or groups of operators shall calculate a risk rate for the activity based on the disturbance regime in the activity area associated with the forest type planned under the activity. For that purpose, the return period of natural disturbance events in the activity area and for the forest type of the activity shall be calculated as the average interval between occurrences of a natural disturbance event. Natural disturbance events shall include at least forest fires, windthrows, and disease and insect outbreaks.

To that end, the datasets on the prediction of the return period, made available by the Commission<sup>23</sup>, may be used. Alternatively, statistics on natural disturbance events covering a period of at least 30 years preceding the start of the activity period may be used to calculate the ratio between the number of natural disturbance events and the given time-period. In the latter case, such ratio shall be estimated using one or more of relevant empirical records, properly documented local surveys, Earth observation datasets, and peer reviewed literature. The area used to calculate return period statistics shall be large enough to allow for the quantification of robust return period statistics for disturbance regimes representative of the geographical region, ecological conditions or administrative level (NUTS2 or NUTS1).

<sup>22</sup> <https://data.jrc.ec.europa.eu/collection/id-00445>

<sup>23</sup> Ibidem

The disturbance regime indicator shall be classified as follows:

- (1) low, where the return period is higher than or equal to 30 years;
- (2) medium, where the return period is between 15 and 30 years;
- (3) high, where the return period is below 15 years.

In the event that a specific tree species is not yet present in the activity area, the disturbance regime indicator shall be based on the general disturbance regime in the activity area.

Based on the combination of forest type and the disturbance regime indicator, the risk rate (expressed as the share of carbon at risk of being released) shall be determined in accordance with Table 7.

Table 7

|                     |                    | Disturbance regime |               |             |
|---------------------|--------------------|--------------------|---------------|-------------|
|                     |                    | <i>Low</i>         | <i>Medium</i> | <i>High</i> |
| <b>Forest type</b>  | <i>Broadleaves</i> | 8 %                | 17 %          | 48 %        |
|                     | <i>Conifers</i>    | 10 %               | 20 %          | 50 %        |
|                     | <i>Mixed</i>       | 8 %                | 17 %          | 45 %        |
| <b>Agroforestry</b> |                    | 4 %                | 8 %           | 24 %        |

Groups of operators may use the weighted average of the risk rate across the activity area.

Where risk mitigation practices beyond the eligibility and minimum sustainability requirements are put in place, such as co-planting of vegetation inhibiting pest infestation, establishment of fire breaks and fire towers, ready access to adequate fire-fighting equipment, the risk rate may be reduced by multiplying it by a factor ranging between 0,8 and 1. Certification schemes may provide additional guidance on these risk mitigation practices and the corresponding factor.

#### 4.2. Liability mechanisms

For activities that generate a temporary net carbon removal benefit, a liability mechanism shall be used.

To that end, operators or groups of operators shall either participate in a buffer pool managed by a certification scheme or obtain and maintain sufficient coverage under an insurance policy or comparable guarantee products. The insurance company or comparable guarantee shall fulfil the relevant financial market rules and shall manage a pool of units.

After each re-certification audit, certification schemes or insurance companies shall allocate to the buffer pool or pool of units a share of units that is equal to the risk rate established in the risk assessment referred to in Sections 4.1.1 and 4.1.3. Where, following an update of the risk assessment, the risk rate is decreased, this shall result in a decreased contribution share to the buffer pool that shall diminish proportionally as the update of the risk assessment approaches the end of the activity.

The certification scheme or insurance company shall ensure that, in the event of an unavoidable and avoidable reversal of the carbon underlying units that have been retired, these units are replaced by an equal number of units from the buffer pool or the pool of units. Such units shall be valid for the same period of time as the units that have been retired. Under an insurance policy or comparable guarantee product, such replacement may not take the form of monetary compensation.

For avoidable reversals for which units have been cancelled from the buffer pool or pool of units, operators shall be fully liable for replenishing the buffer pool or pool of units with an equivalent number of units of the same duration as the units that have been cancelled.

Early termination of a carbon farming sequestration activity shall be considered equal to a full reversal. For soil emission reduction activities, certification schemes shall establish liability mechanisms to deter early termination of the activity.

Certification schemes or insurance companies shall ensure the resilience, sufficiency, and solvency of the buffer pool or pool of units, including through regular stress-testing of the sufficiency of risk coverage. Certification schemes or insurance companies shall publish annually the composition of the buffer pool or pool of units, including the share of units allocated to the pool by vintage, region, and type of carbon farming activity. The stress test shall assess the resilience of the pool to a range of reversal risk scenarios based on, inter alia, the range of risk ratings as well as significant loss events, affecting the activities linked to the pool. The stress test shall occur at least every five years and in the event of a significant loss event. Where necessary based on the results of the stress test, certification schemes or insurance companies shall undertake measures to address risks affecting the resilience, sufficiency and solvency of the buffer pool or pool of units.

After the end of the monitoring period, the buffered units shall expire.

## **5. SUSTAINABILITY**

### **5.1. Minimum sustainability requirements**

During the activity period, operators shall comply with the following minimum sustainability requirements:

#### **(a) climate change mitigation**

Operators may not convert forest land to cropland or grassland, nor convert grassland to cropland in the area under their operational control.

With the exception of peat present in composted biowaste or used as growing media for agroforestry seedlings or for tree nurseries, peat or peat-containing products shall not be used.

#### **(b) climate change adaptation**

The activity shall be consistent with local, sectoral, regional and/or national adaptation strategies and plans, in accordance with national law.

#### **(c) sustainable use and protection of water and marine resources.**

Environmental degradation risks related to preserving water quality and avoiding water stress shall be identified and addressed in accordance with Directive 2000/60/EC of the European Parliament and of the Council<sup>24</sup> and with a river basin management plan, developed in accordance with that Directive.

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<sup>24</sup> Directive 2000/60/EC of the European Parliament and of the Council of 23 October 2000 establishing a framework for Community action in the field of water policy (OJ L 327, 22.12.2000, .p. 1, ELI: <http://data.europa.eu/eli/dir/2000/60/oj>).

The activity shall not hamper the achievement of good environmental status of marine waters nor deteriorate marine waters that are already in good environmental status as defined in Article 3, point 5, of Directive 2008/56/EC of the European Parliament and of the Council<sup>25</sup>.

Where the activity has obtained a development consent addressing the risk of environmental degradation following an environmental impact assessment in accordance with Directive 2011/92/EU of the European Parliament and of the Council<sup>26</sup> which includes an assessment of the impact on water or marine waters in accordance with Directive 2000/60/EC or Directive 2008/56/EC, respectively, no additional assessment of impact on water is required.

- (d) transition to a circular economy, including the efficient use of sustainably sourced bio-based materials

No minimum requirements apply.

- (e) pollution prevention and control

For agriculture and agroforestry on mineral soils and for practices relating to the rewetting and restoration of peatlands and of other organic soils referred to in Section 1.1.2.1, second paragraph, the activity shall comply with the mandatory measures set out in national action programmes established under Council Directive 91/676/EEC<sup>27</sup>.

The amount of applied plant protection products (PPP) shall not increase on average during the activity period as compared to the continuation of PPP application recorded during the reference period that is used to calculate the baseline.

For the practices relating to the rewetting and restoration of peatlands and of other organic soils referred to in Section 1.1.2.1, first paragraph, and for afforestation, the activity may only use pesticides where it is needed to control large-scale outbreaks of pests, diseases and invasive alien species. In such cases alternative approaches or techniques, such as non-chemical alternatives to pesticides are favoured, in accordance with Directive 2009/128/EC. The activity shall not use fertilisers or manure, except, for afforestation, where they are necessary to ensure the survival and growth of young plants, such as in areas affected by desertification, or where specific nutrient imbalances detrimental to tree health are present. The activity shall comply with Regulation (EU) 2019/1009 of the European Parliament and of the Council<sup>28</sup> and with the relevant national implementing law on active ingredients.

- (f) protection and restoration of biodiversity and ecosystems, including soil health as well as avoidance of land degradation

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<sup>25</sup> Directive 2008/56/EC of the European Parliament and of the Council of 17 June 2008 establishing a framework for community action in the field of marine environmental policy (Marine Strategy Framework Directive) (OJ L 164, 25.6.2008, p. 19, ELI: <http://data.europa.eu/eli/dir/2008/56/oj>).

<sup>26</sup> Directive 2011/92/EU of the European Parliament and of the Council of 13 December 2011 on the assessment of the effects of certain public and private projects on the environment (OJ L 26, 28.1.2012, p. 1, ELI: <http://data.europa.eu/eli/dir/2011/92/oj>).

<sup>27</sup> Council Directive 91/676/EEC of 12 December 1991 concerning the protection of waters against pollution caused by nitrates from agricultural sources (OJ L 375, 31.12.1991, p. 11, ELI: <http://data.europa.eu/eli/dir/1991/676/oj>).

<sup>28</sup> Regulation (EU) 2019/1009 of the European Parliament and of the Council of 5 June 2019 laying down rules on the making available on the market of EU fertilising products and amending Regulations (EC) No 1069/2009 and (EC) No 1107/2009 and repealing Regulation (EC) No 2003/2003 (OJ L 170, 25.6.2019, p. 1, ELI: <http://data.europa.eu/eli/reg/2019/1009/oj>).

In areas designated by the national competent authority for conservation or in habitats that are protected, the activity shall be carried out in accordance with the conservation objectives for those areas.

The activity shall not result in the conversion of habitats and species specifically sensitive to biodiversity loss or with high conservation value.

The activity shall exclude the use or release of invasive alien species in accordance with Regulation (EU) No 1143/2014 of the European Parliament and of the Council<sup>29</sup>.

In addition, for practices relating to the rewetting and restoration of peatlands and of other organic soils referred to in Section 1.1.2.1, first paragraph, the activity is in line with applicable restoration objectives and requirements under Union and national law.

For afforestation, the activity shall not result in the deterioration of the conservation status of habitats specifically sensitive to biodiversity loss or with high conservation value or of areas designated for the restoration of such habitats in accordance with national law and EU law. The activity shall not result in substantial deterioration of deadwood nor in substantial soil degradation.

## **5.2. Co-benefits for the protection and restoration of biodiversity and ecosystems, including soil health and the avoidance of land degradation**

Operators or groups of operators may use one or more of the following options to demonstrate compliance with the obligation to generate co-benefits for the protection and restoration of biodiversity and ecosystems, including soil health and the avoidance of land degradation and report such co-benefits:

- (a) the practice has been demonstrated in peer-reviewed scientific literature including synthesis papers, meta reviews or meta-analyses, such as, where applicable, the Commission Evidence library<sup>30</sup>, to have beneficial impacts for biodiversity protection or restoration;
- (b) the activity corresponds to at least one typology of measures referred to in point 14.4 of the Annex to Commission Implementing Regulation (EU) 2025/912<sup>31</sup> and it actively or passively assists in the ecosystem ‘restoration’ as defined in Article 3, point (3), of Regulation (EU) 2024/1991 of the European Parliament and of the Council<sup>32</sup>;

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<sup>29</sup> Regulation (EU) No 1143/2014 of the European Parliament and of the Council of 22 October 2014 on the prevention and management of the introduction and spread of invasive alien species (OJ L 317, 4.11.2014, p. 35, ELI: <http://data.europa.eu/eli/reg/2014/1143/oj>).

<sup>30</sup> [https://datam.jrc.ec.europa.eu/datam/mashup/JRC\\_FP\\_EVIDENCE\\_LIBRARY/index.html](https://datam.jrc.ec.europa.eu/datam/mashup/JRC_FP_EVIDENCE_LIBRARY/index.html)

<sup>31</sup> Commission Implementing Regulation (EU) 2025/912 of 19 May 2025 laying down rules for the application of Regulation (EU) 2024/1991 of the European Parliament and of the Council as regards a uniform format for the national restoration plan (OJ L, 2025/912, 20.5.2025, ELI: [http://data.europa.eu/eli/reg\\_impl/2025/912/oj](http://data.europa.eu/eli/reg_impl/2025/912/oj)). The typologies of measures are available at <https://biodiversity.europa.eu/europes-biodiversity/nature-restoration/reference-portal-for-nature-restoration-regulation/documentation>

<sup>32</sup> Regulation (EU) 2024/1991 of the European Parliament and of the Council of 24 June 2024 on nature restoration and amending Regulation (EU) 2022/869 (OJ L, 2024/1991, 29.7.2024, ELI: <http://data.europa.eu/eli/reg/2024/1991/oj>).

- (c) the activity is aligned with management plans and other implementing measures under Directive 2009/147/EC of the European Parliament and of the Council<sup>33</sup> and Council Directive 92/43/EEC<sup>34</sup> which correspond to the ecological requirements of the natural habitats as described in Article 6 of Directive 92/43/EEC or has been qualitatively demonstrated to contribute to improved condition or maintain good condition of a habitat type or a habitat of species of Union interest, using established methodologies such as those under those Directives, or the Union-wide methodology on mapping and assessing ecosystem condition for other ecosystems<sup>35</sup>;
- (d) for afforestation, in addition to points (a), (b) and (c) the following practices are considered to deliver co-benefits for the protection and restoration of biodiversity and ecosystems, including soil health and the avoidance of land degradation:
  - (i) afforestation with at least three tree species whereby each used tree species covers at least 20 % of the total species composition, in a way that species composition is maintained throughout the monitoring period;
  - (ii) maintenance of pioneering species in open forest and bare soil areas;
  - (iii) keeping of trees damaged by wind and of deadwood on site;
  - (iv) promotion of natural regeneration;
  - (v) maintenance of riparian buffer zones along streams in the activity area.

### 5.3. Other sustainability co-benefits

#### 5.3.1. Demonstration of other sustainability co-benefits

Operators or groups of operators may report other co-benefits for sustainability objectives. Those co-benefits may be demonstrated through either:

- (a) an activity-based approach: the implementation of a specific practice is assumed to result in the co-benefit;
- (b) a result-based approach: the co-benefit is quantified by showing improvements against relevant sustainability indicators.

Where the activity-based approach referred to in the first paragraph, point (a), is used, the practice shall have been demonstrated in peer-reviewed scientific literature to have beneficial impacts for the relevant sustainability objectives when carried out in similar pedo-climatic conditions. In this respect, where applicable, the Commission Evidence library may be used.

Where a result-based approach referred to in the first paragraph, point (b), is used, the sustainability indicators shall be consistent with applicable indicators and methodologies under national and Union legislation, such as Directive (EU) 2025/2360 of the European Parliament and of the Council<sup>36</sup> or Regulation (EU) 2024/1991.

Certification schemes shall enable knowledge sharing on the approaches used to demonstrate the co-benefits.

<sup>33</sup> Directive 2009/147/EC of the European Parliament and of the Council of 30 November 2009 on the conservation of wild birds (OJ L 20, 26/01/2010, p. 7, ELI: <http://data.europa.eu/eli/dir/2009/147/oj>).

<sup>34</sup> Council Directive 92/43/EEC of 21 May 1992 on the conservation of natural habitats and of wild fauna and flora (OJ L 206, 22.7.1992, p. 7, ELI: <http://data.europa.eu/eli/dir/1992/43/oj> ).

<sup>35</sup> Vallecillo et al., 2022, <https://data.europa.eu/doi/10.2760/13048>

<sup>36</sup> Directive (EU) 2025/2360 of the European Parliament and of the Council of 12 November 2025 on soil monitoring and resilience (Soil Monitoring Law) (OJ L, 2025/2360, 26.11.2025, ELI: <http://data.europa.eu/eli/dir/2025/2360/oj>).

### 5.3.2. *Examples of sustainability co-benefits from agriculture and agroforestry on mineral soils*

The following non-exhaustive list includes examples of co-benefits for sustainability objectives:

- (a) climate change mitigation, beyond the temporary net carbon removal benefit and the net soil emission reduction benefit generated through the activity, resulting in:
  - (i) a reduction in GHG emissions related to the on-farm use of machinery and energy;
  - (ii) a reduction in GHG emissions related to the production of fertilisers;
- (b) climate change adaptation, including:
  - (i) improved water retention;
  - (ii) prevention of soil compaction and erosion;
- (c) sustainable use and protection of water and marine resources, including:
  - (i) improved water use efficiency and increased water availability;
  - (ii) improved infiltration towards the groundwater and improved water quality;
- (d) transition to a circular economy, including the efficient use of sustainably sourced bio-based materials, such as the use of:
  - (i) fewer resources or certified reused and recycled products;
  - (ii) fertilising products made of recovered materials;
- (e) pollution prevention and control, including a reduction in air, soil, or water pollution;
- (f) protection and restoration of biodiversity and ecosystems including soil health, as well as the avoidance of land degradation, resulting in:
  - (i) an improvement in the grassland butterfly index;
  - (ii) an improvement in the share of agricultural land with high-diversity landscape feature;
  - (iii) an improvement in the farmland bird index;
  - (iv) an improvement in soil ecological status.

### 5.3.3. *Examples of sustainability co-benefits from the rewetting and restoration of peatlands and of other organic soils*

The following non-exhaustive list includes examples of co-benefits for sustainability objectives:

- (a) climate change mitigation, beyond the net soil emission reductions generated through the activity, resulting in:
  - (i) a reduction in GHG emissions related to the on-farm use of machinery and energy;

- (ii) a reduction in methane peak emissions resulting from the rewetting<sup>37</sup>;
- (b) climate change adaptation, including:
  - (i) increased evaporative cooling and increased thermal conductivity of the soil/peat;
  - (ii) disaster risk management, through compliance with the technical screening criteria set out in Section 3.1. of Annex I to Commission Delegated Regulation (EU) 2023/2486<sup>38</sup>;
- (c) sustainable use and protection of water and marine resources, including:
  - (i) improved water availability and water use efficiency;
  - (ii) increased water retention and decreased flood risk;
  - (iii) improved water quality;
  - (iv) increased base flow;
- (d) transition to a circular economy, including the efficient use of sustainably sourced bio-based materials, such as the use of:
  - (i) biomass from paludiculture;
  - (ii) dead plant parts as organic fertiliser, compost, or crop residues;
  - (iii) recovered materials;
- (e) pollution prevention and control, including:
  - (i) if the land before rewetting was in agricultural use, the prevention or reduction in harmful phosphorus emissions to the surface waters;
  - (ii) a reduction in air, soil or water pollution;
  - (iii) avoidance in soil compaction;
- (f) protection and restoration of biodiversity and ecosystems including soil health, as well as the avoidance of land degradation, resulting in the following:
  - (i) an improvement in the grassland butterfly index;
  - (ii) an improvement in the share of agricultural land with high-diversity landscape features;
  - (iii) an improvement in the farmland bird index;

<sup>37</sup> By means of example, through practices included in the Ramsar Technical Report – Convention on Wetlands. (2021). Global guidelines for peatland rewetting and restoration. Ramsar Technical Report No. 11. Gland, Switzerland: Secretariat of the Convention on Wetlands.

<sup>38</sup> Commission Delegated Regulation (EU) 2023/2486 of 27 June 2023 supplementing Regulation (EU) 2020/852 of the European Parliament and of the Council by establishing the technical screening criteria for determining the conditions under which an economic activity qualifies as contributing substantially to the sustainable use and protection of water and marine resources, to the transition to a circular economy, to pollution prevention and control, or to the protection and restoration of biodiversity and ecosystems and for determining whether that economic activity causes no significant harm to any of the other environmental objectives and amending Commission Delegated Regulation (EU) 2021/2178 as regards specific public disclosures for those economic activities (OJ L, 2023/2486, 21.11.2023, ELI: [http://data.europa.eu/eli/reg\\_del/2023/2486/oj](http://data.europa.eu/eli/reg_del/2023/2486/oj)).

- (iv) with respect to paludiculture, an enhancement of habitat structure, species richness, and ecosystem function.

#### 5.3.4. *Examples of sustainability co-benefits from afforestation*

The following non-exhaustive list includes examples of co-benefits for sustainability objectives:

- (a) climate mitigation, beyond the temporary net carbon removal benefit and the net soil emission reduction benefit generated through the activity, resulting in:
  - (i) a reduction in GHG emissions from non-use of fertilisers, where this is not mandatory under national legislation;
  - (ii) a reduction in GHG emissions from use of peat-free saplings;
  - (iii) a reduction in GHG emissions related to the use of machinery and energy;
  - (iv) carbon removals from deadwood and/or litter;
- (b) climate change adaptation, including:
  - (i) improved flood prevention and water circulation;
  - (ii) microclimate regulation;
  - (iii) improved resilience;
- (c) sustainable use and protection of water and marine resources, including:
  - (i) improved water quality;
  - (ii) improved water retention and decreased flood risk;
  - (iii) marine ecosystem protection;
- (d) transition to a circular economy, including the efficient use of sustainably sourced bio-based materials, including:
  - (i) energy recovery of any burning natural (organic) materials and other suitable waste;
  - (ii) use of certified reusable, recyclable or recycled products, e.g. plant pots, anti-bite sleeves or anti-bite fences;
- (e) pollution prevention and control, including:
  - (i) a limitation in waste generation;
  - (ii) a reduction in air, soil or water pollution;
- (f) protection and restoration of biodiversity and ecosystems, including soil health and the avoidance of land degradation, resulting in the following:
  - (i) an improvement in standing deadwood;
  - (ii) an improvement in lying deadwood;
  - (iii) an improvement in the share of forests with uneven-aged structure;
  - (iv) an improvement in forest connectivity;
  - (v) an improvement in the share of forests dominated by native tree species;
  - (vi) an improvement in tree species diversity;
  - (vii) an improvement in the common forest bird index.

